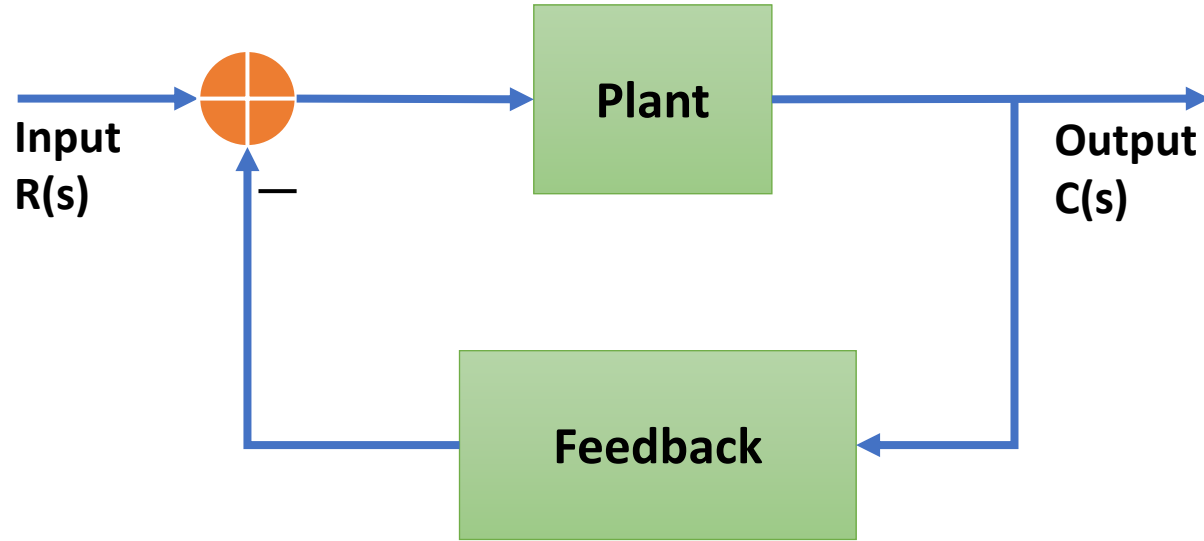


Compensators

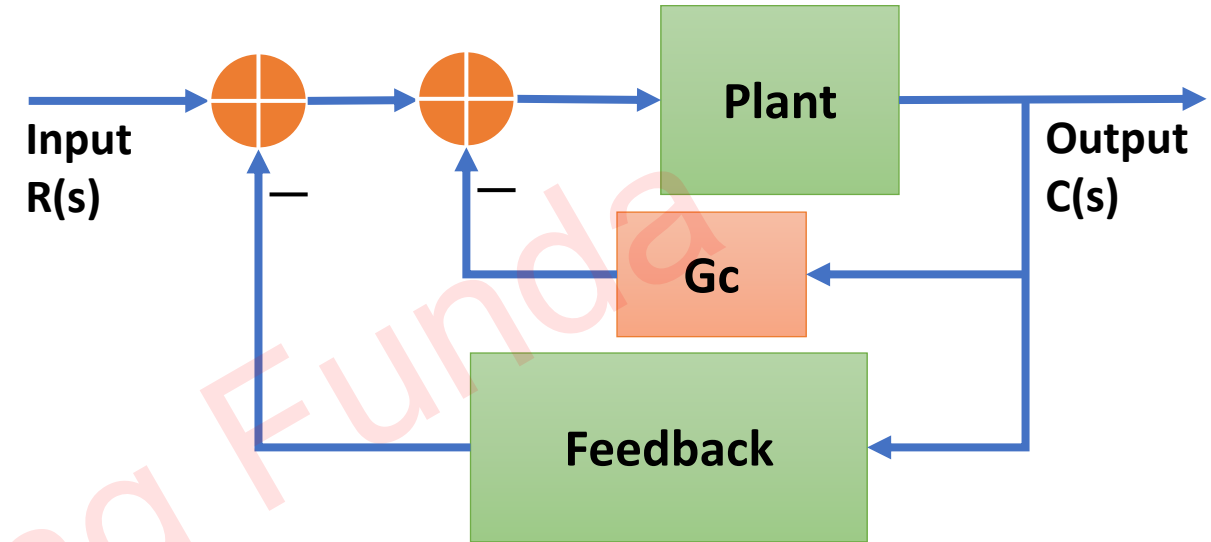
Basics of Compensators

- ❑ Compensators are used to modify the dynamics of a system to achieve specific performance requirements. Common objectives include:
 - Improving system stability.
 - Reducing overshoot or settling time in transient responses.
 - Enhancing steady-state accuracy.
 - Increasing the bandwidth of the system for better response to changes.
 - Shaping the system's frequency response.
- ❑ Time Domain Specifications: Rise Time, Maximum Peak Overshoot, Settling Time, and Damping Ratio. For Time Domain Specifications, Root Locus Technique is used in Designing the System.
- ❑ Frequency Domain Specifications: Phase Margin, Gain Margin, Resonant Peak, and Bandwidth. For Frequency Domain Specifications, The Bode Plot Technique is used in Designing the System.

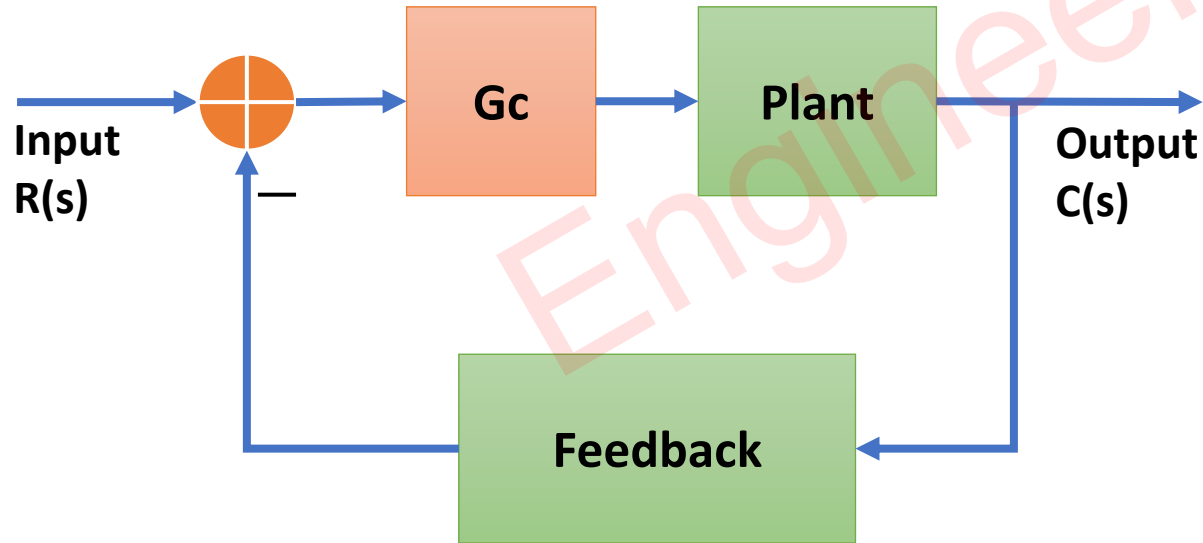
Types of Compensators



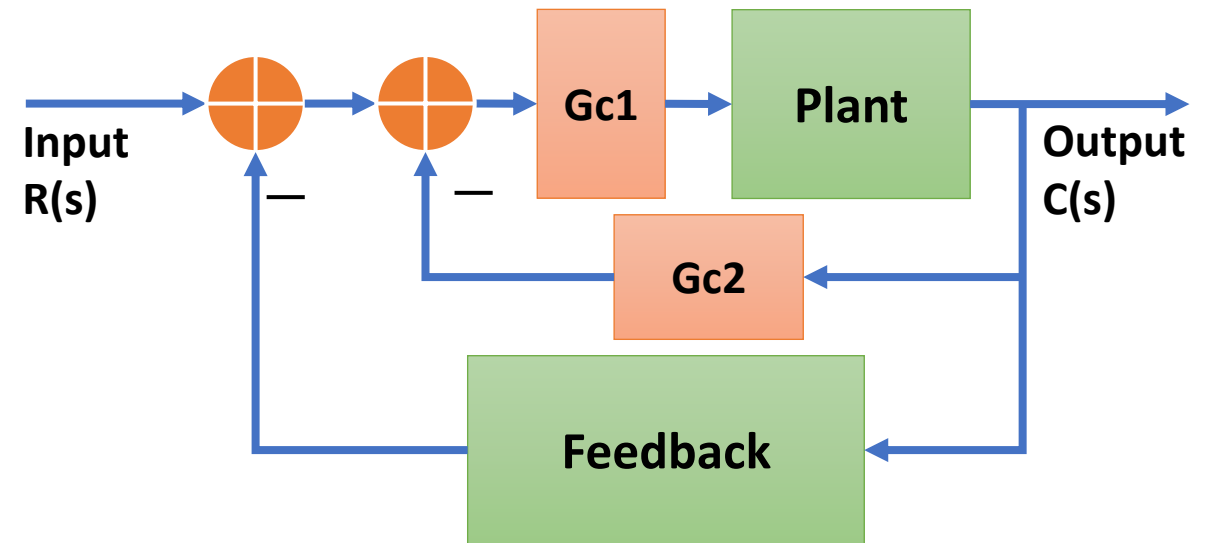
❖ Feedback Compensator {Parallel Compensator}



❖ Cascade Compensator {Series Compensator}



❖ Cascade with Feedback Compensator



Classifications of Compensators

❑ Lead Compensator

- **Purpose:** Improves Transient Response and Increase System Stability
- **Effect:** Adds Lead Phase (Positive Phase) to a specific frequency range.
- **Application:** Speeds up the response and Increases the Gain Margin.

❑ Lag Compensator

- **Purpose:** Improves Steady State Error without affecting Transient Performance.
- **Effect:** Adds Lag Phase (Negative Phase) to a specific frequency range.
- **Application:** Enhances Low-Frequency Gain and Improves Steady State Error.

❑ Lead-Lag Compensator

- **Purpose:** Combines the advantages of Lead and Lag Compensators.
- **Effect:** Improves Transient Response with Steady State Response.
- **Application:** Used in Complex Systems for multiple aspects.

Lag Compensator

Outlines

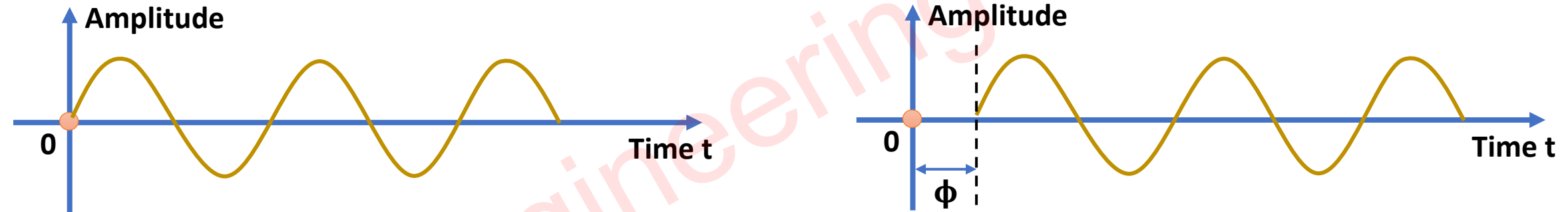
- ❑ Basics of Lag Compensator
- ❑ S-Plane Representation of Lag Compensator
- ❑ Maximum Phase Condition of Lag Compensator
- ❑ Lag Compensator using Electrical Network
- ❑ Bode Plot of Lag Compensator

Engineering Funda

Lag Compensator

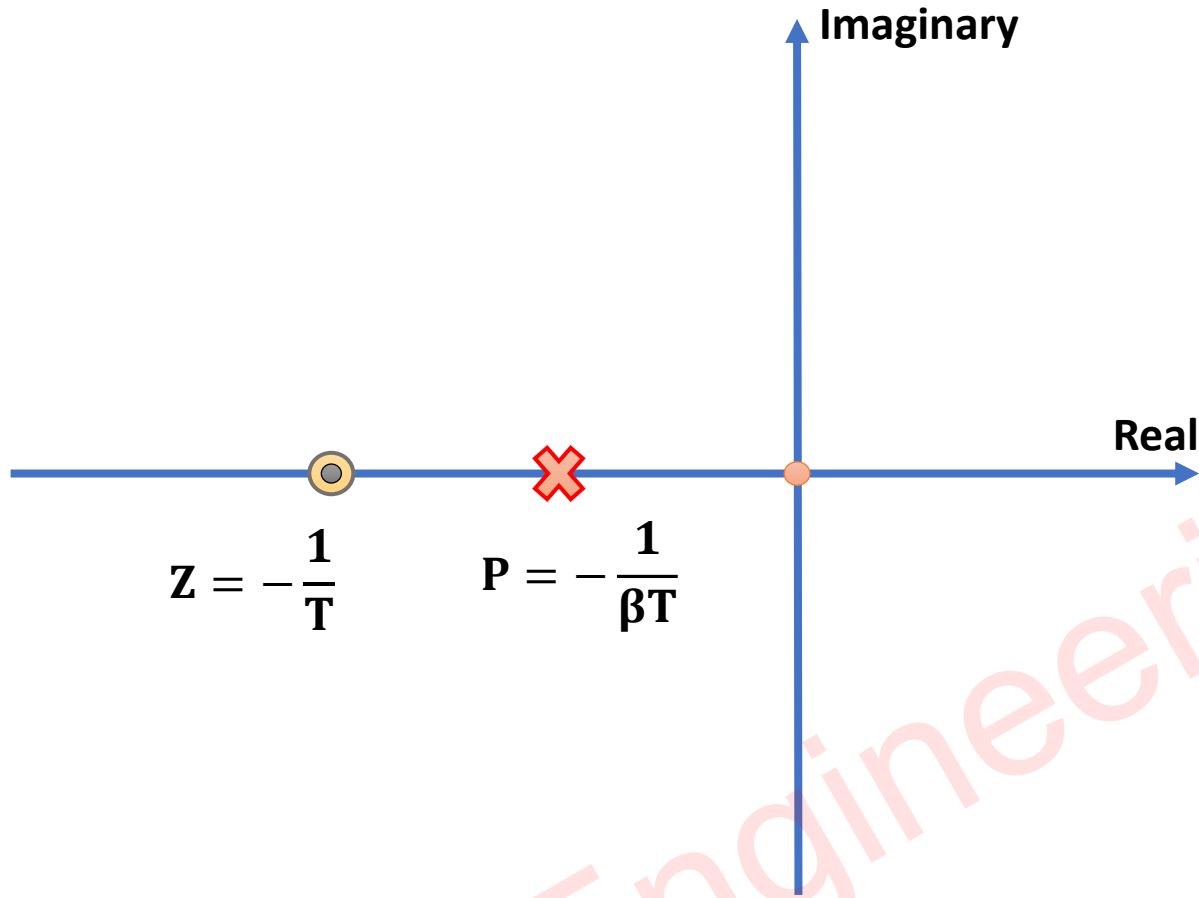
Basics of Lag Compensator

❑ A Compensator with the lag Network characteristics is called a Lag Compensator.



❑ Lag Compensator is a Low Pass Filter. High-frequency noise signals are attenuated.

S-Plane Representation of Lag Compensator



□ Transfer Function of a Lag Compensator.

$$\Rightarrow G_c(s) = \frac{1}{\beta} \left(\frac{s + \frac{1}{T}}{s + \frac{1}{\beta T}} \right)$$

▪ Here, $\beta > 1$

Maximum Phase Condition of Lag Compensator

→ Put $s = j\omega$ in $K_c(s)$.

$$\Rightarrow K_c(j\omega) = \frac{1}{\beta} \left(\frac{j\omega + \frac{1}{T}}{j\omega + \frac{1}{\beta T}} \right)$$

→ phase

$$\Rightarrow \phi = +\tan^{-1}(\omega T) - \tan^{-1}(\omega \beta T)$$

→ magnitude

$$\Rightarrow m = \frac{1}{\beta} \left(\frac{\sqrt{\frac{1}{T^2} + \omega^2}}{\sqrt{\frac{1}{\beta^2 T^2} + \omega^2}} \right)$$

→ For maximum phase, $\frac{d\phi}{d\omega} = 0$

$$\Rightarrow \frac{T}{1 + \omega^2 T^2} - \frac{\beta T}{1 + \omega^2 \beta^2 T^2} = 0$$

$$\Rightarrow 1 + \omega^2 \beta^2 T^2 = \beta + \omega^2 T^2 \beta$$

$$\Rightarrow 1 - \beta = \omega^2 T^2 \beta - \omega^2 \beta T^2$$

$$\Rightarrow \cancel{1} - \beta = \omega^2 T^2 \beta (\cancel{1} - \beta)$$

$$\Rightarrow \boxed{\omega_m^2 = \frac{1}{T\sqrt{\beta}}}$$

$$\Rightarrow \text{Put } \omega_m = \frac{1}{T\sqrt{\beta}} \text{ into } \phi = \tan^{-1}(\omega T) - \tan^{-1}(\omega \beta T)$$

$$\Rightarrow \phi_m = \tan^{-1}\left(\frac{1}{\sqrt{\beta}}\right) - \tan^{-1}(\sqrt{\beta})$$

$$= \tan^{-1}\left(\frac{\frac{1}{\sqrt{\beta}} - \sqrt{\beta}}{1 + \frac{1}{\sqrt{\beta}}\sqrt{\beta}}\right)$$

$$\Rightarrow \boxed{\phi_m = \tan^{-1}\left(\frac{1 - \beta}{2\sqrt{\beta}}\right)}$$

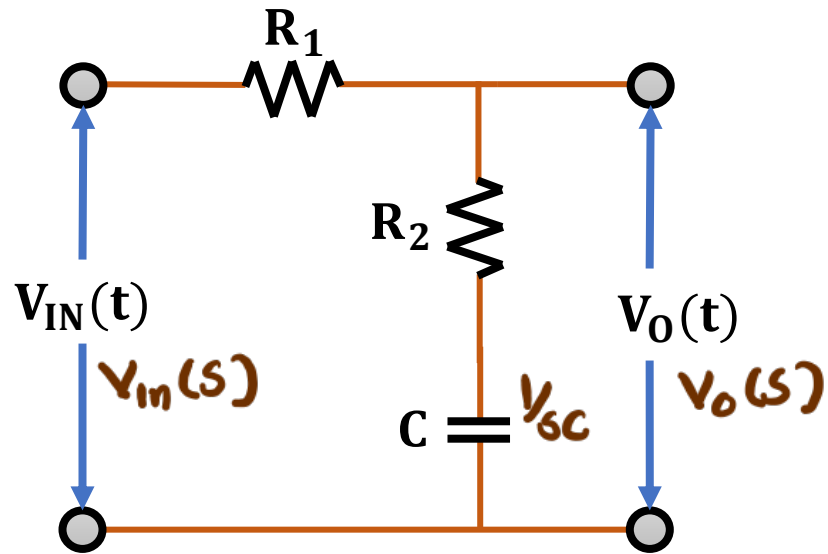
$$\Rightarrow \text{Put } \omega_m = \frac{1}{T\sqrt{\beta}} \text{ into } m = \frac{1}{\beta} \frac{\sqrt{\omega^2 + \gamma_T^2}}{\sqrt{\omega^2 + \gamma_{\beta T}^2}}$$

$$- m = \frac{1}{\beta} \frac{\sqrt{\frac{1}{T^2\beta} + \gamma_T^2}}{\sqrt{\frac{1}{T^2\beta} + \gamma_{\beta T}^2}}$$

$$= \frac{\sqrt{\beta}}{\beta}$$

$$\boxed{m = \frac{1}{\sqrt{\beta}}}$$

Lag Compensator using Electrical Network



⇒ Apply voltage divider rule.

$$\Rightarrow V_O(s) = V_{in}(s) \left(\frac{R_2 + 1/sC}{R_1 + R_2 + 1/sC} \right)$$

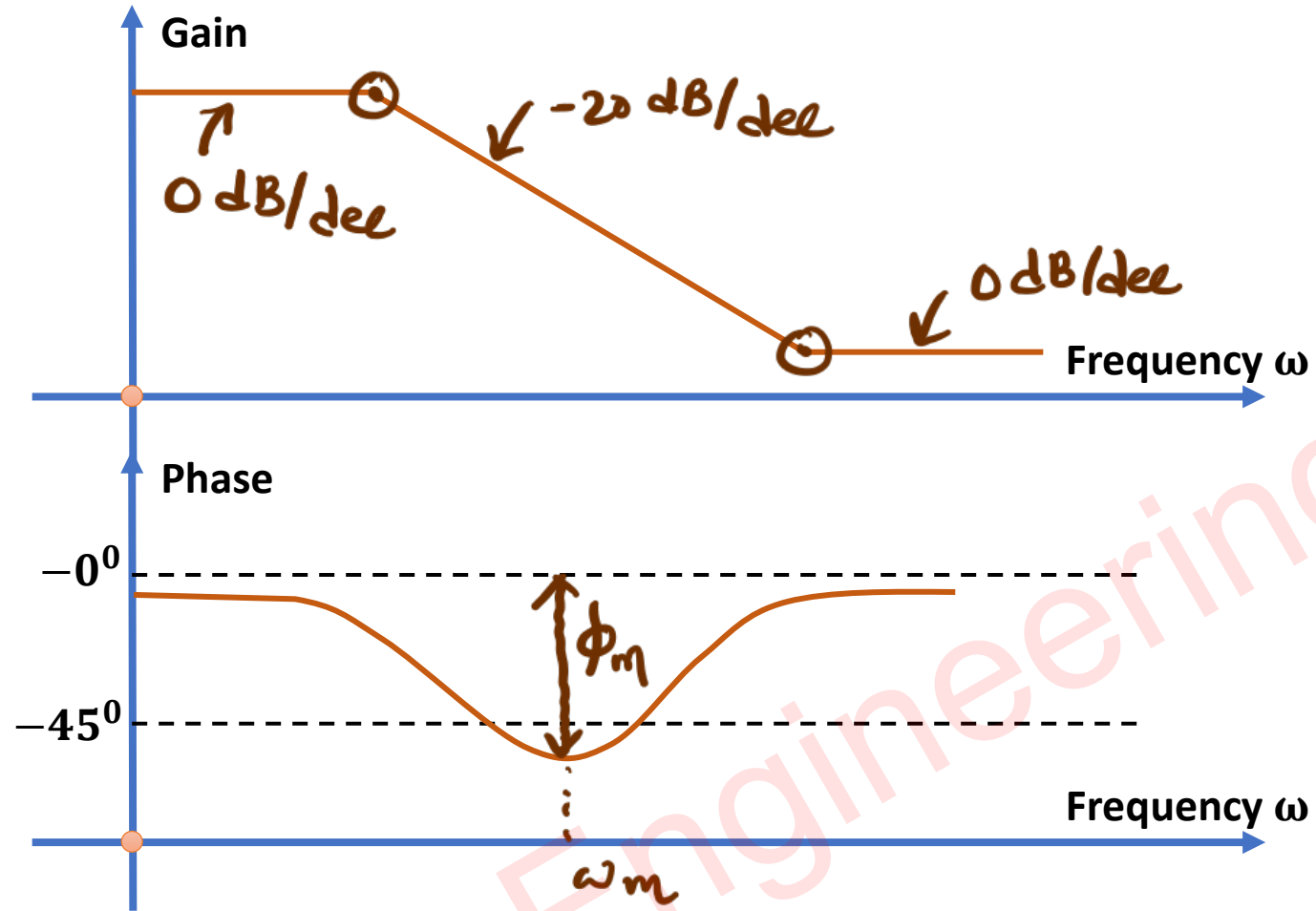
$$\Rightarrow \frac{V_O(s)}{V_{in}(s)} = G_c(s) = \frac{1 + R_2 sC}{1 + (R_1 + R_2)sC}$$

$$\Rightarrow G_c(s) = \frac{R_2 C}{(R_1 + R_2)C} \left[\frac{s + 1/R_2 C}{s + 1/(R_1 + R_2)C} \right]$$

$$\Rightarrow G_c(s) = \frac{R_2}{R_1 + R_2} \left[\frac{s + 1/R_2 C}{s + \frac{1}{R_2 C} \left(\frac{R_2}{R_1 + R_2} \right)} \right]$$

$$- \beta = \frac{R_1 + R_2}{R_2}, \quad T = R_2 C$$

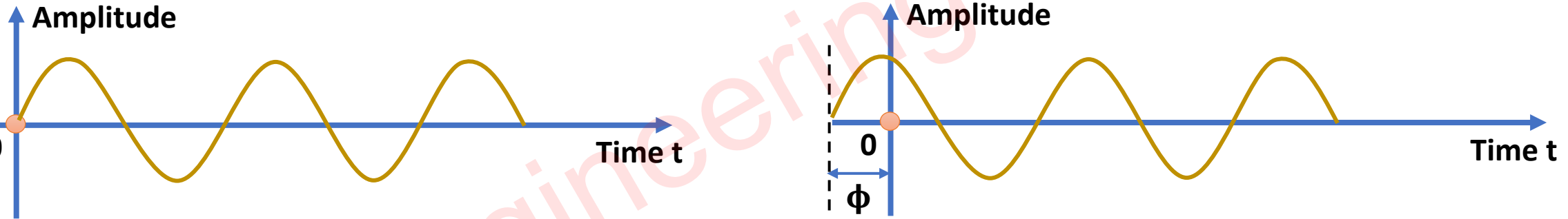
Bode Plot of Lag Compensator ($P < Z$)



Lead Compensator

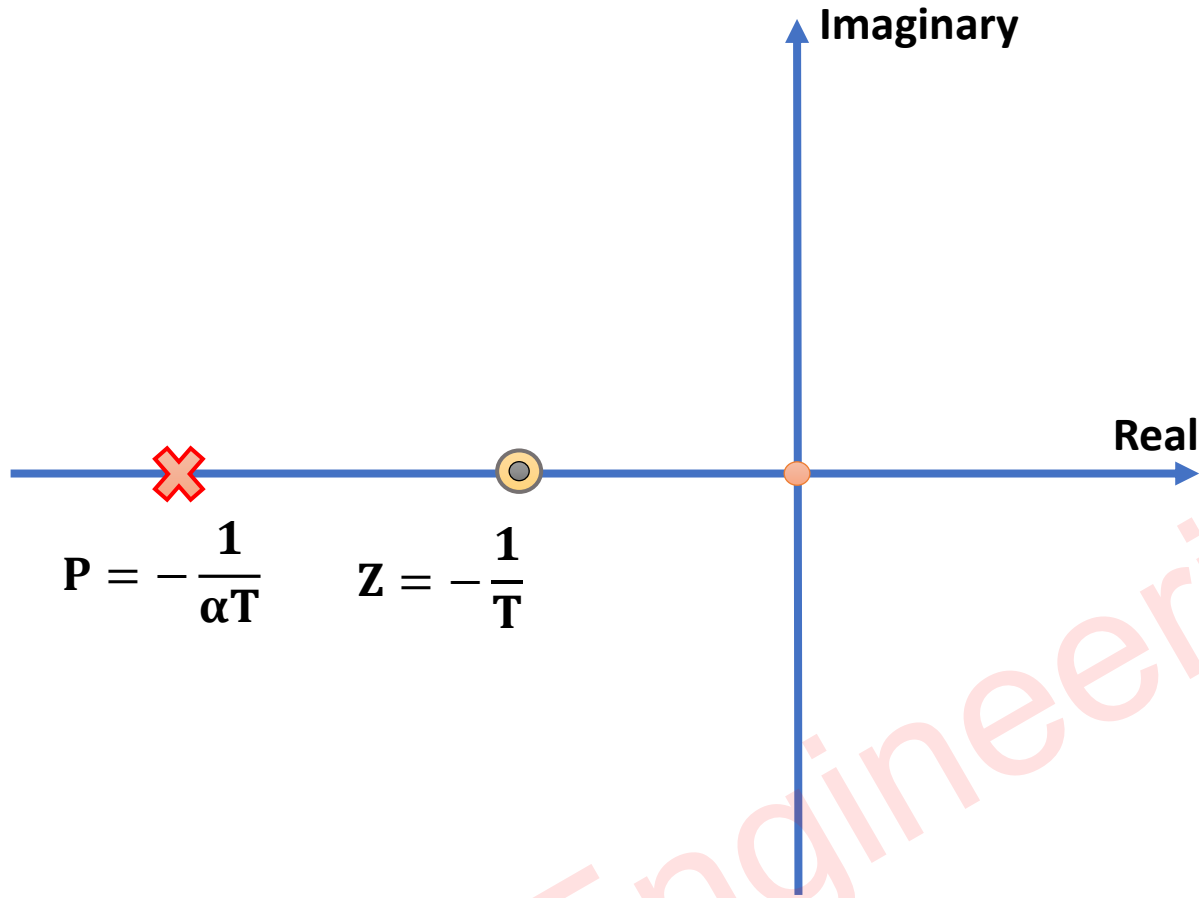
Basics of Lead Compensator

❑ A Compensator with the Lead Network characteristics is called a Lead Compensator.



❑ Lead Compensator is a High Pass Filter. Low-frequency noise signals are attenuated and High-Frequency signals are amplified.

S-Plane Representation of Lead Compensator



□ Transfer Function of a Lead Compensator.

$$\Rightarrow G_c(S) = \frac{1}{\alpha} \left(\frac{S + \frac{1}{T}}{S + \frac{1}{\alpha T}} \right)$$

▪ Here, $\alpha < 1$

Maximum Phase Condition of Lead Compensator

⇒ Put $s = j\omega$ in $G_c(s)$.

$$\Rightarrow G_c(j\omega) = \frac{1}{\alpha} \left(\frac{j\omega + \frac{1}{T}}{j\omega + \frac{1}{\alpha T}} \right), \alpha < 1.$$

⇒ phase ϕ

$$\Rightarrow \phi = \tan^{-1}(\omega T) - \tan^{-1}(\omega \alpha T)$$

⇒ Magnitude m

$$\Rightarrow m = \frac{1}{\alpha} \frac{\sqrt{\omega^2 + \frac{1}{T^2}}}{\sqrt{\omega^2 + \frac{1}{\alpha^2 T^2}}}$$

⇒ To get maximum phase ϕ_m , $\frac{d\phi}{d\omega} = 0$

$$\Rightarrow \frac{T}{1 + \omega^2 T^2} - \frac{\alpha T}{1 + \omega^2 \alpha^2 T^2} = 0$$

$$\Rightarrow 1 + \omega^2 \alpha^2 T^2 = \alpha + \alpha \omega^2 T^2$$

$$\Rightarrow 1/\alpha = \omega^2 T^2 (\alpha (1/\alpha))$$

$$\Rightarrow \omega^2 = \frac{1}{T^2 \alpha}$$

$$\Rightarrow \omega_m = \frac{1}{T\sqrt{\alpha}}$$

$\leadsto$ To get Maximum phase put $\omega_m = \frac{1}{T\sqrt{\alpha}}$ into ϕ

$$\Rightarrow \phi_m = \tan^{-1} \left(\frac{1}{\sqrt{\alpha}} \right) - \tan^{-1} (\sqrt{\alpha})$$

$$= \tan^{-1} \left(\frac{\frac{1}{\sqrt{\alpha}} - \sqrt{\alpha}}{1 + \frac{1}{\alpha} \alpha} \right)$$

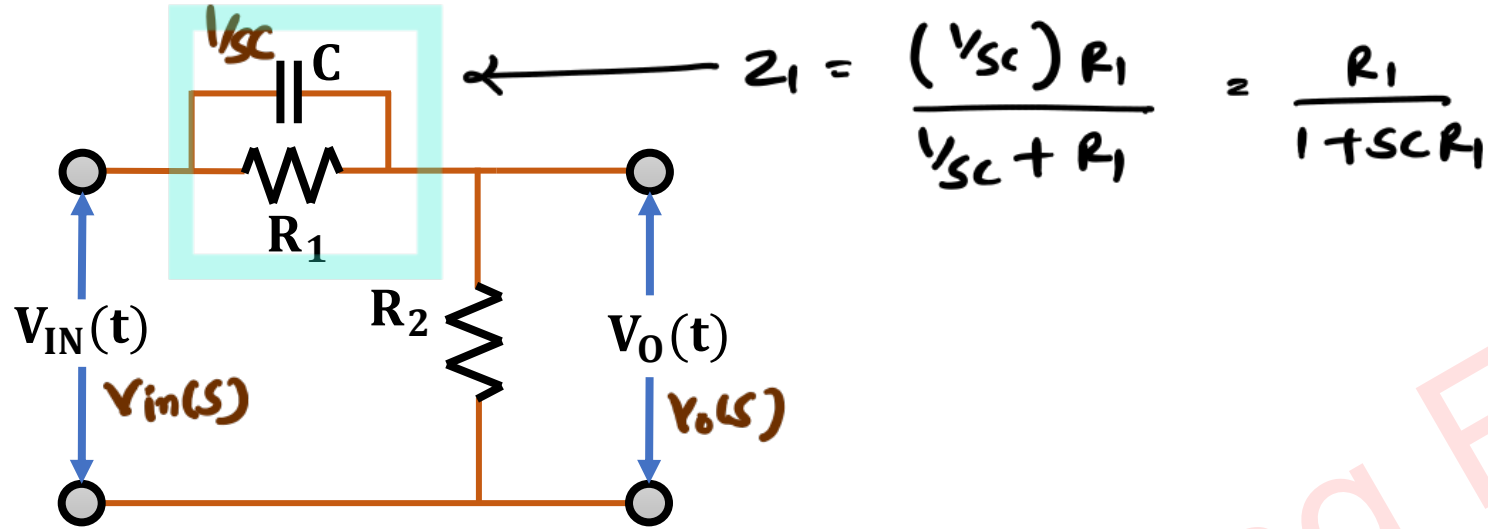
$$\Rightarrow \phi_m = \tan^{-1} \left(\frac{1-\alpha}{2\sqrt{\alpha}} \right)$$

$\leadsto$ To get Magnitude at max. phase, put $\omega_m = \frac{1}{T\sqrt{\alpha}}$ into M .

$$\Rightarrow M = \frac{1}{\alpha} \frac{\sqrt{\frac{1}{T^2 \alpha} + 1}}{\sqrt{\frac{1}{T^2 \alpha} + \frac{1}{\alpha} T^2}}$$

$$\Rightarrow M = \frac{\sqrt{\alpha}}{\alpha} = \boxed{\frac{1}{\sqrt{\alpha}}}$$

Lead Compensator using Electrical Network



⇒ Apply Voltage Divider Rule.

$$\Rightarrow V_o(s) = V_{in}(s) \left(\frac{R_2}{Z_1 + R_2} \right)$$

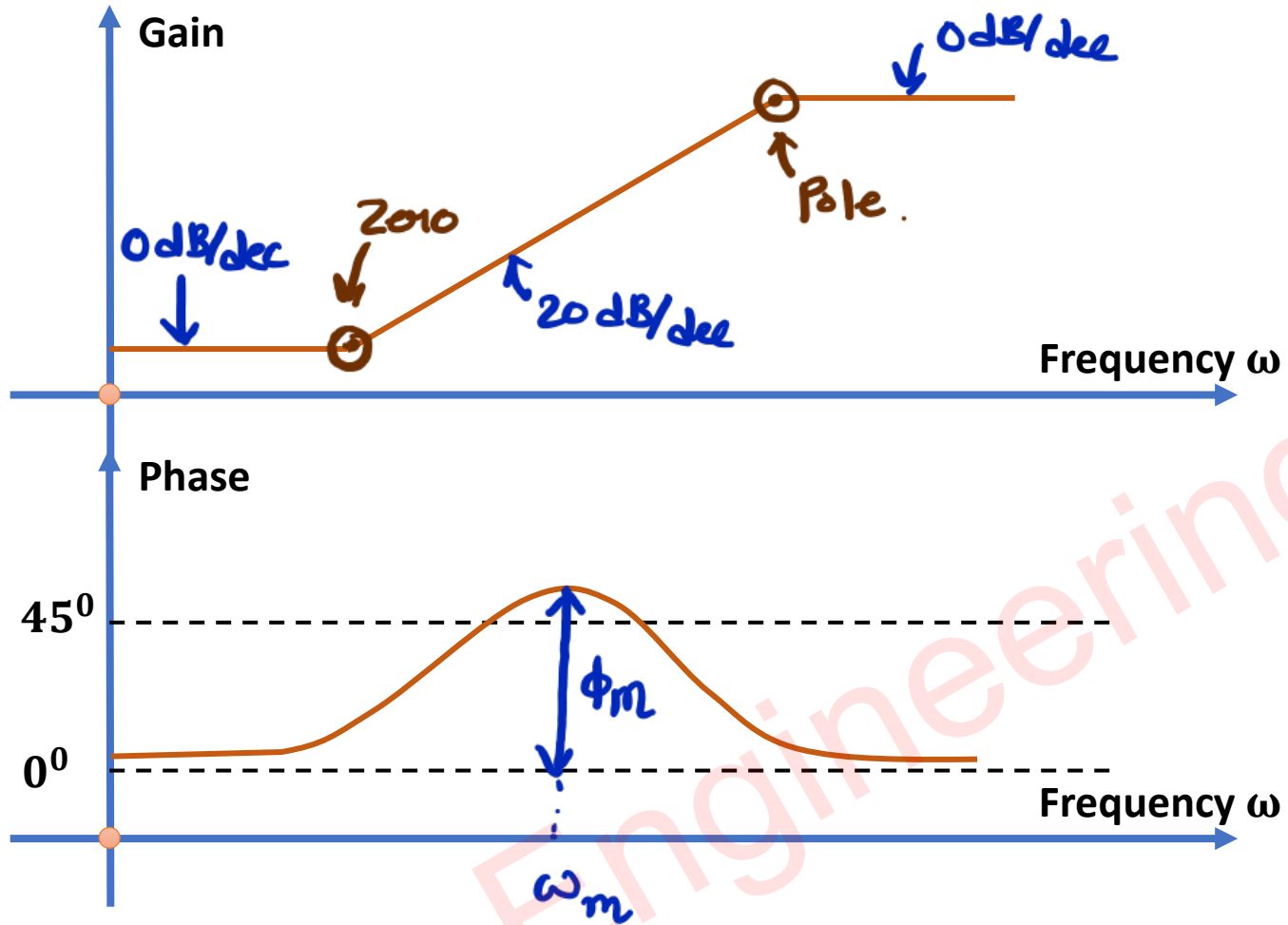
$$\Rightarrow \frac{V_o(s)}{V_{in}(s)} = \frac{R_2}{\frac{R_1}{1 + sC R_1} + R_2} = \frac{R_2 (1 + sC R_1)}{R_1 + R_2 + R_1 R_2 sC}$$

$$\Rightarrow \frac{V_o(s)}{V_{in}(s)} = \frac{\cancel{C R_1} R_2 (s + 1/\cancel{C R_1})}{\cancel{C R_1} R_2 (s + \frac{R_1 + R_2}{R_1 R_2 C})} = \frac{s + 1/\cancel{C R_1}}{s + \frac{1}{C R_1} \left(\frac{R_1 + R_2}{R_2} \right)}$$

$$\Rightarrow T = C R_1, \quad \alpha = \frac{R_2}{R_1 + R_2} < 1$$

Engineering Funda

Bode Plot of Lead Compensator ($P_\omega > P_z$)



Lag-Lead Compensator

Outlines

- ❑ Basics of Lag-Lead Compensator
- ❑ S-Plane Representation of Lag-Lead Compensator
- ❑ Transfer Function of Lag-Lead Compensator
- ❑ Lag-Lead Compensator using Electrical Network

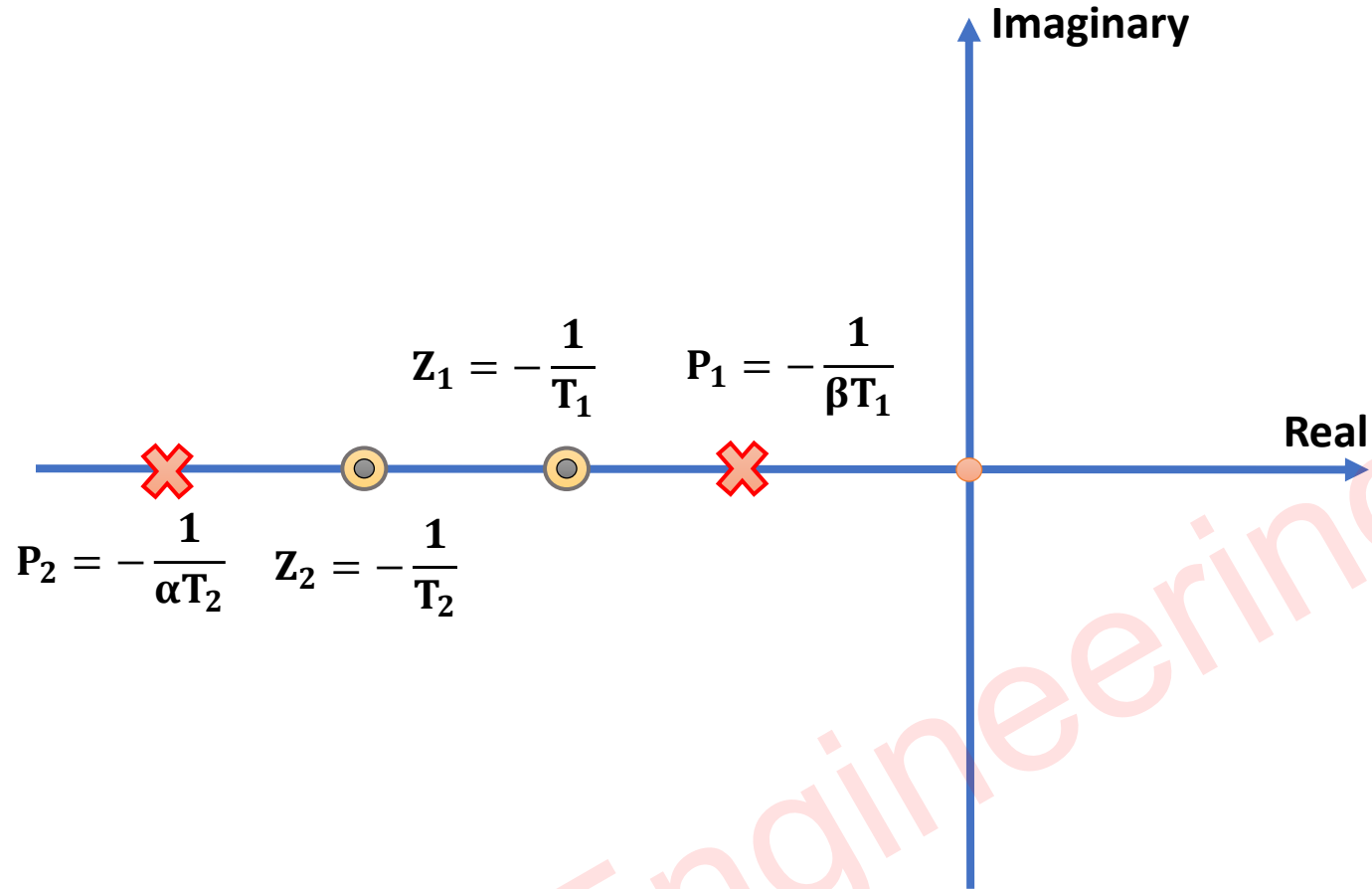
Engineering Funda

Lag-Lead Compensator

Basics of Lag-Lead Compensator

- ❑ Lag-Lead Compensator produces Phase Lag at One Frequency Region and Phase Lead at Another Frequency Region.
- ❑ It Combines the advantages of Lag and Lead Compensators.
- ❑ It is used to Improve Transient Response with Steady State Response.
- ❑ It is used in Complex Systems for the multiple aspects.
- ❑ Rather to use Lag and Lead Compensator separately, it is economical to use Lag-Lead Compensator.

S-Plane Representation of Lag-Lead Compensator



□ Transfer Function of a Lag-Lead Compensator.

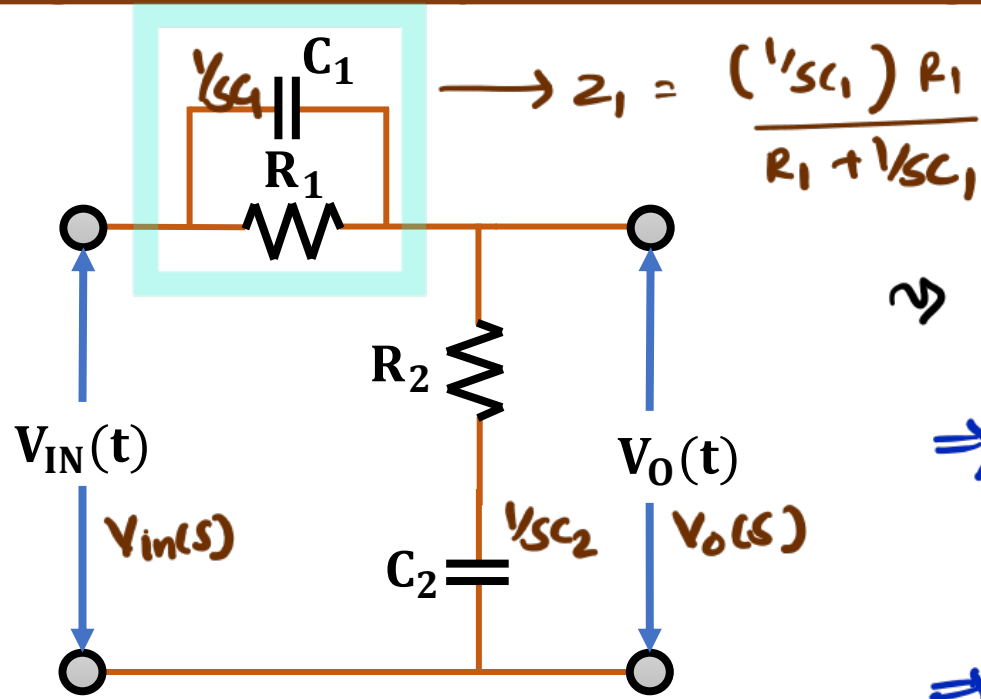
$$\Rightarrow G_c(s) = \left(\frac{s + \frac{1}{T_1}}{s + \frac{1}{\beta T_1}} \right) \left(\frac{s + \frac{1}{T_2}}{s + \frac{1}{\alpha T_2}} \right)$$

▪ Here, $\beta > 1$ and $\alpha < 1$

$$\Rightarrow K_c(s) = \frac{(s + \frac{1}{T_1})(s + \frac{1}{T_2})}{s^2 + s(\frac{1}{\beta T_1} + \frac{1}{\alpha T_2}) + \frac{1}{\alpha \beta T_1 T_2}}$$

$$\Rightarrow K_c(s) = \frac{(s + \frac{1}{T_1})(s + \frac{1}{T_2})}{(s + \frac{\alpha}{T_1})(s + \frac{1}{\alpha T_2})} \quad \text{or} \quad \frac{(s + \frac{1}{T_1})(s + \frac{1}{T_2})}{(s + \frac{1}{\beta T_1})(s + \frac{\beta}{T_2})}$$

Lag-Lead Compensator using Electrical Network



⇒ Apply Voltage divider Rule.

$$\Rightarrow V_O(s) = V_{IN}(s) \left(\frac{R_2 + 1/sC_2}{(R_2 + 1/sC_2) + \frac{R_1 (1/sC_1)}{R_1 + 1/sC_1}} \right)$$

$$\Rightarrow \frac{V_O(s)}{V_{IN}(s)} = \frac{(R_1 + 1/sC_1)(R_2 + 1/sC_2)}{\frac{R_1}{sC_1} + (R_1 + 1/sC_1)(R_2 + 1/sC_2)}$$

$$\Rightarrow \frac{V_O(s)}{V_{IN}(s)} = \frac{(1 + R_1 C_1 s)(1 + R_2 C_2 s)}{R_1 s C_2 + (1 + R_1 C_1 s)(1 + R_2 C_2 s)}$$

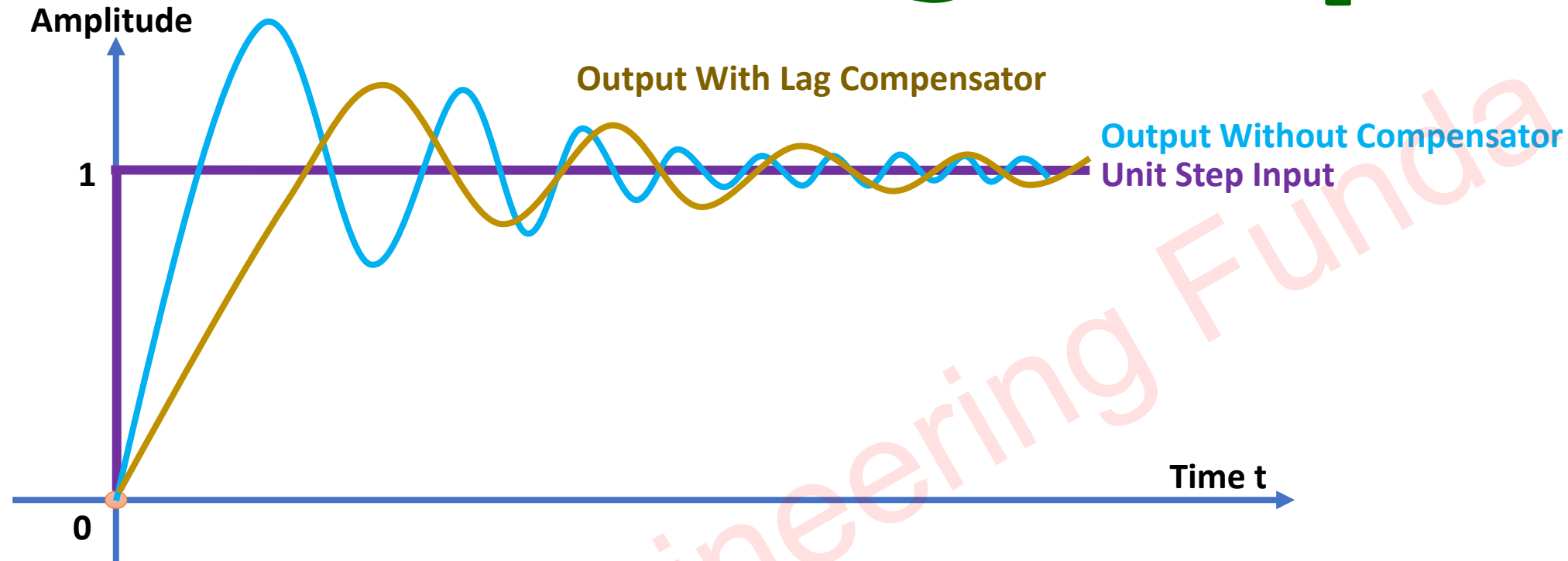
$$\Rightarrow \frac{V_O(s)}{V_{IN}(s)} = \frac{(1 + R_1 C_1 s)(1 + R_2 C_2 s)}{s^2 R_1 R_2 C_1 C_2 + s(R_1 C_2 + R_1 C_1 + R_2 C_2) + 1}$$

$$\Rightarrow \frac{V_o(s)}{V_{in}(s)} = \frac{(s + \frac{1}{R_1 C_1})(s + \frac{1}{R_2 C_2})}{s^2 + s\left(\frac{1}{R_2 C_1} + \frac{1}{R_2 C_2} + \frac{1}{R_1 C_1}\right) + \frac{1}{R_1 R_2 C_1 C_2}}$$

$$\sim T_1 = R_1 C_1, \quad T_2 = R_2 C_2, \quad \alpha \beta = 1$$

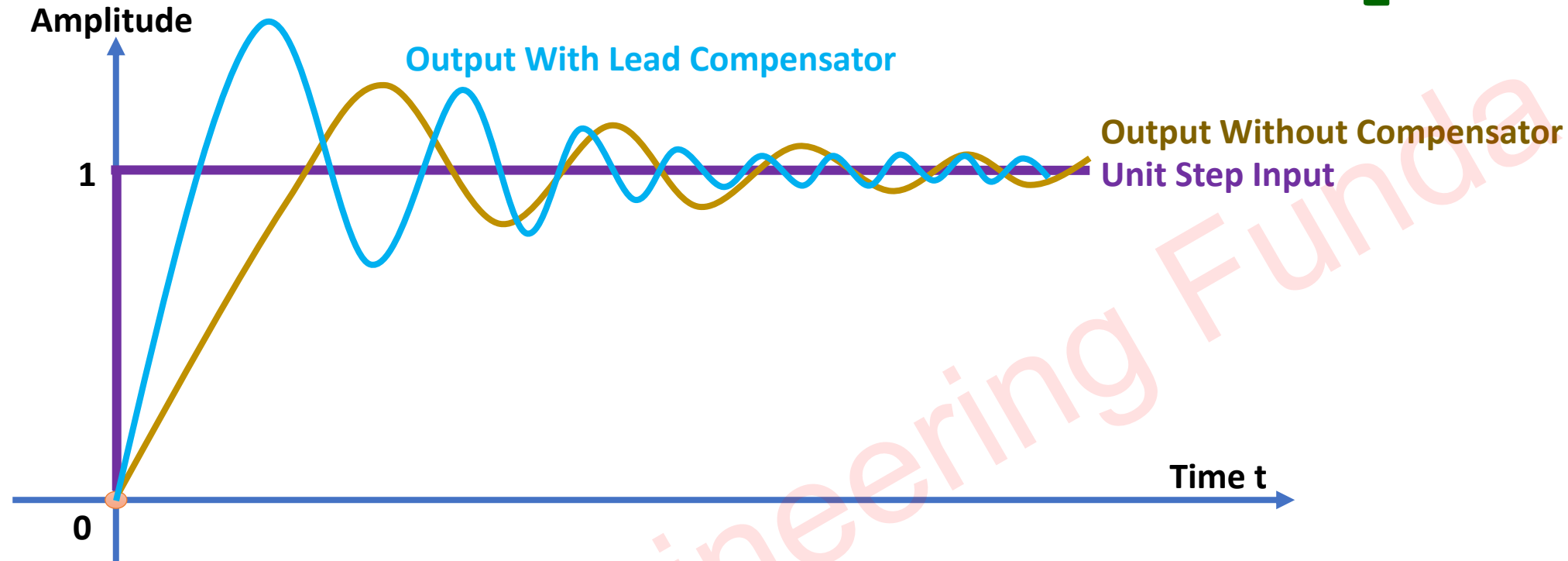
$$(\alpha < 1, \beta > 1)$$

Effects of Lag Compensator



- ❑ Increased Rise Time and Settling Time {Slower Transient Response}.
- ❑ Decrease in Bandwidth and Maximum Peak Overshoot.
- ❑ Improved Steady State Response.
- ❑ It will Eliminate High Frequency Noise.

Effects of Lead Compensator



- ❑ Reduced Rise Time and Settling Time {Improved & Faster Transient Response}.
- ❑ Increase in Bandwidth and Maximum Peak Overshoot.
- ❑ Reduced Gain Cross Over Frequency. Hence, there is an increase in Gain and Phase Margin.
Hence, there is an Improved Stability.
- ❑ High-frequency Noise may get entered.
- ❑ Steady State Error is not affected.

Examples of Compensator

1. Transfer Function of the Compensator is given by:

$$\Rightarrow D(S) = \frac{0.5S + 1}{0.05S + 1}$$

Find the type and Maximum Phase of the given Compensator.

$$\begin{aligned}\Rightarrow D(S) &= \frac{1}{K} \left(\frac{S + 1/T}{S + 1/KT} \right) \\ &= \frac{0.5}{0.05} \left(\frac{S + 1/0.5}{S + 1/(0.5)(0.1)} \right) \\ &= \frac{1}{0.1} \left(\frac{S + 1/0.5}{S + 1/(0.5)(0.1)} \right)\end{aligned}$$

$$\begin{aligned}\Rightarrow K &= 0.1, \quad T = 0.5 \\ &= \alpha\end{aligned}$$

$\Rightarrow$ Lead Compensator

$$\begin{aligned}\rightarrow \phi_m &= \tan^{-1} \left(\frac{1-\alpha}{2\sqrt{\alpha}} \right) \\ &= \tan^{-1} \left(\frac{1-0.1}{2\sqrt{0.1}} \right) \\ &= 54.9^\circ \\ \rightarrow \omega_m &= \frac{1}{T\sqrt{\alpha}} = \frac{1}{0.5\sqrt{0.1}} \\ &= 6.32 \text{ rad/sec}\end{aligned}$$

2. Transfer Function of the Compensator is given by:

$$\Rightarrow D(S) = \frac{1 + 0.2S}{1 + 2S}$$

Find the type of the given Compensator.

$$\begin{aligned}\Rightarrow D(S) &= \frac{1}{K} \left(\frac{S + \frac{1}{T}}{S + \frac{1}{KT}} \right) \\ &= \frac{0.2}{2} \left(\frac{S + \frac{1}{0.2}}{S + \frac{1}{(0.2)(10)}} \right) \\ &= \frac{1}{10} \left(\frac{S + \frac{1}{0.2}}{S + \frac{1}{(0.2)(10)}} \right)\end{aligned}$$

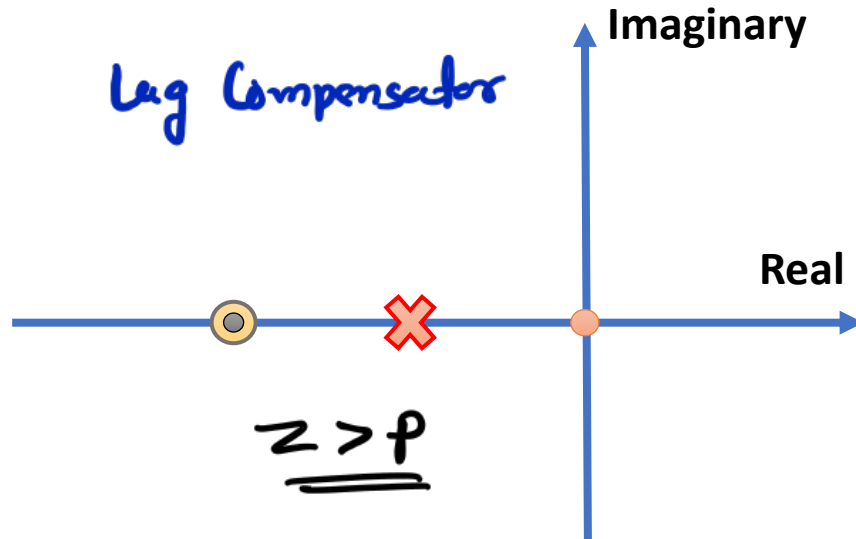
$$- K = 10 > 1, \beta = 10$$

Lag Compensator

3. The pole-zero configuration of the phase lead compensator is given by:

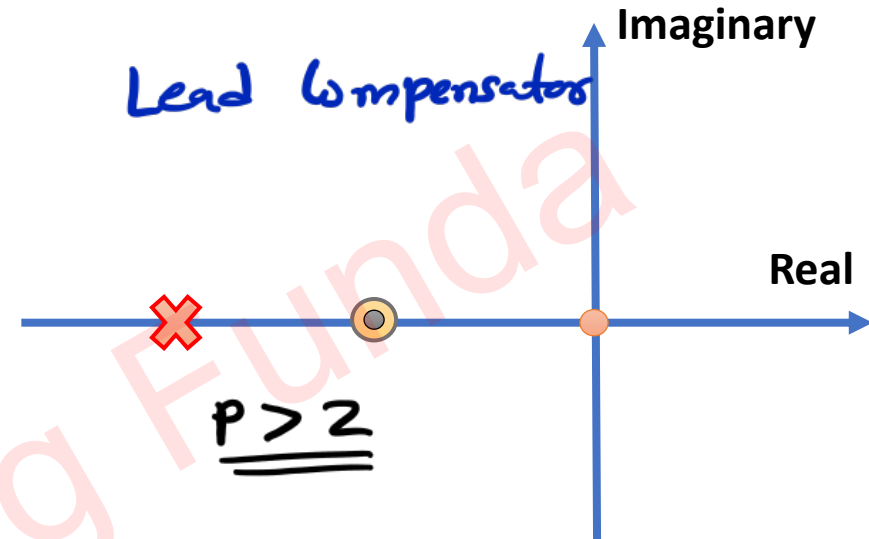
A.

Lag Compensator



B.

Lead Compensator



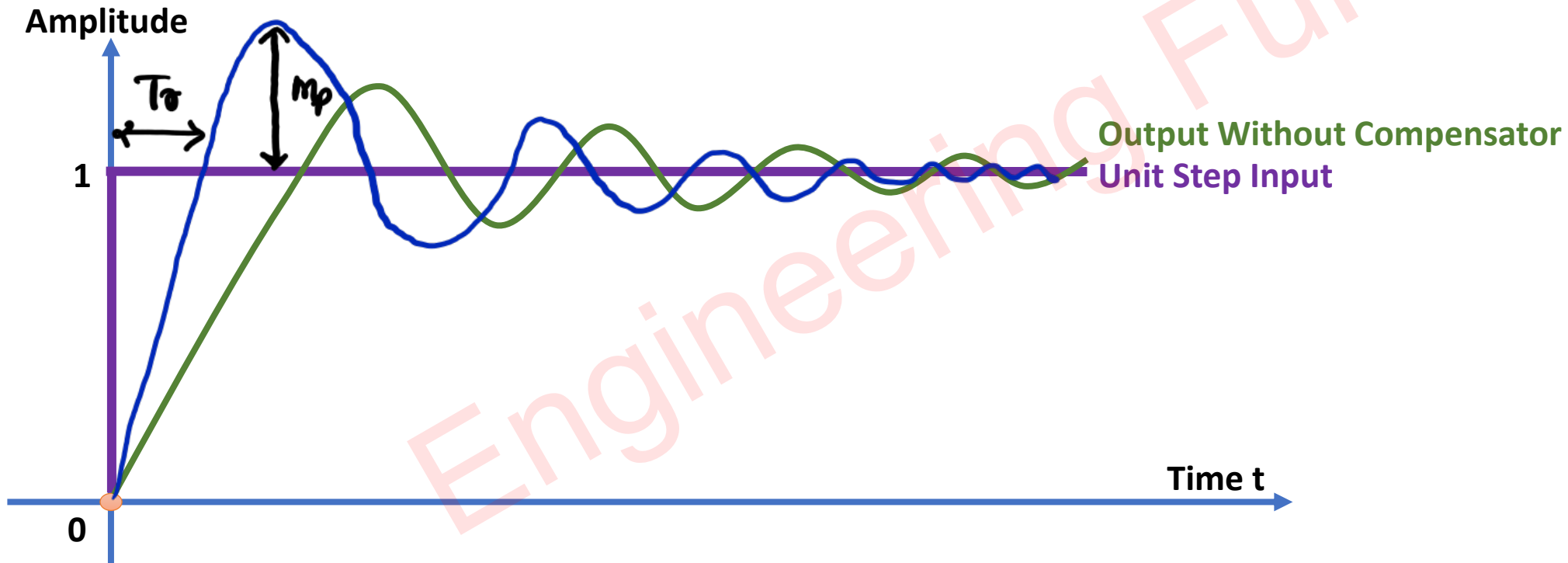
$$- z = \frac{1}{T}, \quad p = \frac{1}{KT}$$

$$\leadsto K > 1 \Rightarrow K = \beta \text{ Lag Compensator}$$

$$\leadsto K < 1 \Rightarrow K = \alpha \text{ Lead Compensator}$$

4. The phase lead compensator is used to

- A. Increase in the rise time and decrease in the overshoot
- ✓ B. decrease in the rise time and Increase in the overshoot
- C. Increase in the rise time and in the overshoot
- D. decrease in the rise time and in the overshoot



5. A Lead Compensator used for closed-loop controller has the following transfer function:

$$\Rightarrow D(S) = k \frac{[1 + S/a]}{[1 + S/b]}$$

For Such Lead Compensator.

- ✓ A. $a < b$
- B. $b < a$
- C. $a > kb$
- D. $a < kb$

$$\Rightarrow D(S) = \frac{kb}{a} \frac{(s+a)}{(s+b)}$$

- For Lead Compensator

$$- p > z \Rightarrow b > a$$

- For Lag compensator,

$$- z > p \Rightarrow a > b$$

6. The Transfer Function of two compensator are given below:

$$\underline{\text{Lead}} \Rightarrow C_1 = 10 \frac{(S + 1)}{(S + 10)} \Rightarrow C_2 = \frac{(S + 10)}{10(S + 1)} \underline{\text{Lag}}$$

Find the type of the given Compensator.

- $P > Z$

Lead Compensator

- $Z > P$

Lag Compensator

7. The Maximum phase shift that can be obtained by using a lead compensator with a transfer function

$$\Rightarrow G_c(s) = \frac{4(1 + 0.15s)}{(1 + 0.05s)} = K \frac{(s + 1/T)}{(s + 1/2T)}$$

$$= \frac{4 \times 0.15}{0.05} \left(\frac{s + 1/0.15}{s + 1/(0.15)(1/3)} \right)$$

$$- T = 0.15, \quad \alpha = 1/3$$

$$\Rightarrow \phi_m = \tan^{-1} \left(\frac{1-\alpha}{2\sqrt{\alpha}} \right)$$

$$= \tan^{-1} \left(\frac{1-1/3}{2\sqrt{1/3}} \right)$$

$$= 30^\circ$$

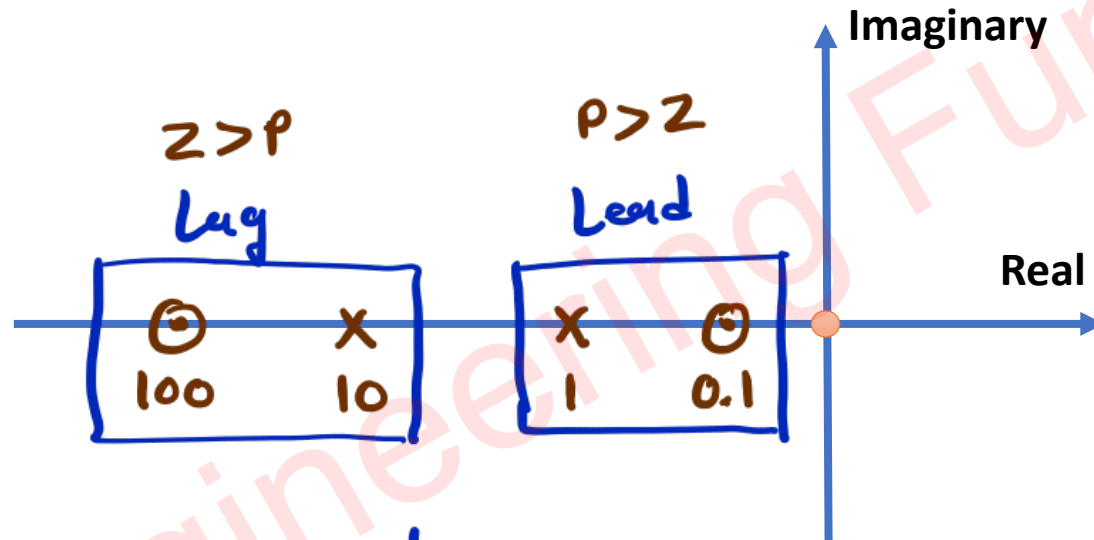
- a. 15 degree
- ✓ b. 30 degree
- c. 45 degree
- d. 60 degree

8. The transfer function $C(S)$ of a compensator is given below:

$$\Rightarrow C(S) = \frac{\left(1 + \frac{S}{0.1}\right) \left(1 + \frac{S}{100}\right)}{(1 + S) \left(1 + \frac{S}{10}\right)}$$

The frequency range in which the phase lead introduced by the compensator reaches the maximum is

- a. ☒ $0.1 < \omega < 1$
- b. $1 < \omega < 10$
- c. $10 < \omega < 100$
- d. $\omega > 100$



- For Lead Compensator

$$1 > \omega > 0.1$$

9. The given transfer function represents a

$$\Rightarrow G_c(S) = \frac{(1 + 0.5S)}{(1 + S)}$$

- ✓ a. Lag Network
- b. Lead Network
- c. Lag-Lead Network
- d. Proportional Controller

$$- z = 2, p = 1$$

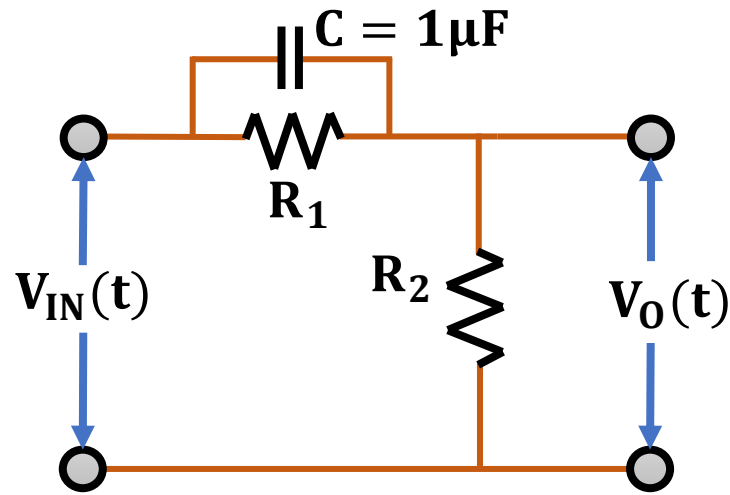
$$- z > p$$

- Lag Compensator.

10. The transfer function of a phase lead network, as shown in the figure below is given by:

$$\Rightarrow G_c(S) = K \frac{(1 + 0.3S)}{(1 + 0.17S)} = K \frac{(S + 1/0.3)}{(S + 1/0.17)} = \frac{K \cdot 0.3}{0.17} \frac{(S + 1/0.3)}{(S + 1/0.17)}$$

Find the values of the Resistance R1 and R2.



$$- T = 0.3$$

$$- \alpha T = 0.17 \Rightarrow \alpha = \frac{0.17}{0.3} = 0.567$$

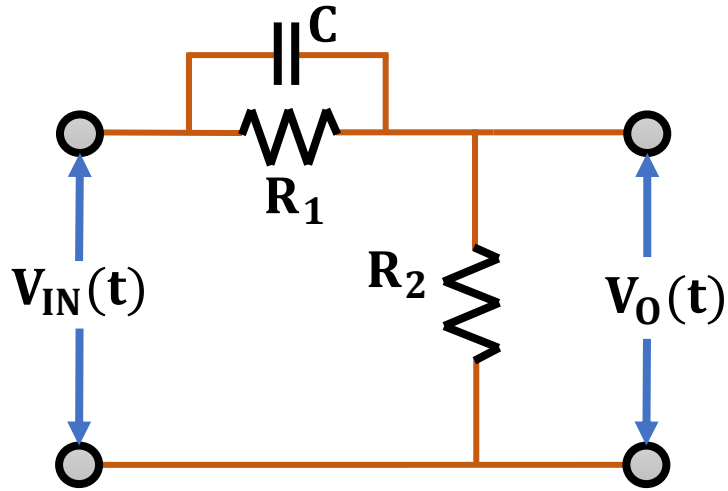
$$\begin{aligned} - T &= R_1 C \Rightarrow R_1 = T/C \\ &= 0.3 / 10^{-6} \\ &= 300 \text{ K}\Omega \end{aligned}$$

$$- T = R_1 C$$

$$- \alpha = \frac{R_2}{R_1 + R_2}$$

$$\begin{aligned} - 0.567 &= \frac{R_2}{300 + R_2} \Rightarrow 0.567 \times 300 + R_2 \cdot 0.567 = R_2 \\ &\Rightarrow R_2 (1 - 0.567) = 0.567 \times 300 \\ &\Rightarrow R_2 = 392 \text{ K}\Omega \end{aligned}$$

11. For the given network identify the maximum phase in terms of R_1 and R_2 . Also, determine the maximum phase in terms of the Sine of Angle.



$$\approx \phi = \tan^{-1} \left(\frac{1-\alpha}{2\sqrt{2}} \right) \Rightarrow \tan \phi = \frac{1-\alpha}{2\sqrt{2}}$$

$$\Rightarrow \alpha = \frac{R_2}{R_1 + R_2}$$

$$\Rightarrow \tan \phi = \frac{R_1 / (R_1 + R_2)}{2 \sqrt{R_2 / (R_1 + R_2)}} = \frac{R_1 \sqrt{R_1 + R_2}}{2 \sqrt{R_2} (R_1 + R_2)} = \frac{R_1}{2 \sqrt{R_2} (R_1 + R_2)}$$

$$= \sqrt{R_1^2 + 4(R_2(R_1 + R_2))}$$

$$= \sqrt{(R_1 + 2R_2)^2}$$

$$= R_1 + 2R_2$$

$$-\sin \phi_m = \frac{R_1}{R_1 + 2R_2}$$